Day 60 – Performance Tuning & Scalability

Objective

Establish measurable performance practices and a horizontal scaling design.

Included utilities

A bounded TTL cache, ordered batch processor and local callable benchmark helper. The benchmark reports mean, median, p95, minimum and maximum latency.

Scaling strategy

Place stateless API workers behind a load balancer; move resume parsing to a durable queue and independently scale worker pools; keep real-time interview requests on a latency-monitored synchronous path; externalize cache and job state for multi-instance deployments; use object storage for documents and a database for metadata.

Load-test plan

Measure p50/p95/p99 latency, throughput, error rate, queue wait, CPU, memory and cache hit rate at increasing concurrency. Test cold/warm cache, batch sizes, large resumes, dependency timeouts and retry storms.

Limitation

No production traffic, model endpoint, distributed cache, queue or load environment was available in this batch. Do not present local microbenchmark numbers as production performance improvements.